06 December 2025 13:37

Strings

In Java, a String is an object that represents a sequence of characters.

Akash => String

A k a s h => sequence of chars

package => java.lang

Strings are immutable => once a object is created, its value cannot be changed.

String → class → Object → Heap memory

Address will be stored in variable inside call stack.

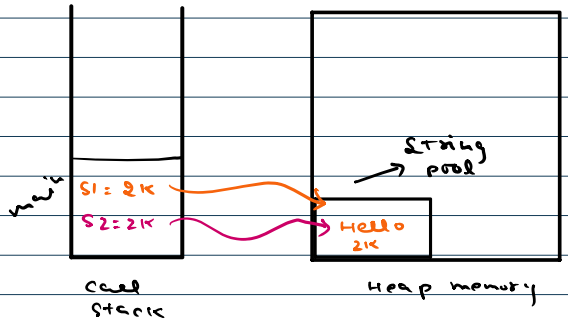
String pool

```
String s1 = "Akash";
String s2 = new String("Akash");
```

→ Heap memory

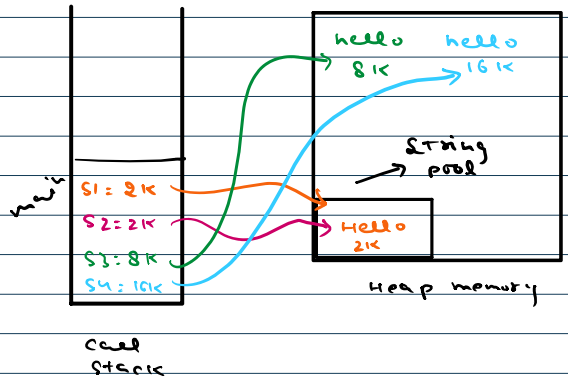
```
public class StringDemo {
    public static void main(String args[]) {
        String s1 = "hello";
        String s2 = "hello";

    }
}
```

JVM




```
public class StringDemo {
    public static void main(String args[]) {
        String s1 = "hello";
        String s2 = "hello";

        String s3 = new String( original: "hello");
        String s4 = new String( original: "hello");

    }
}
```

JVM

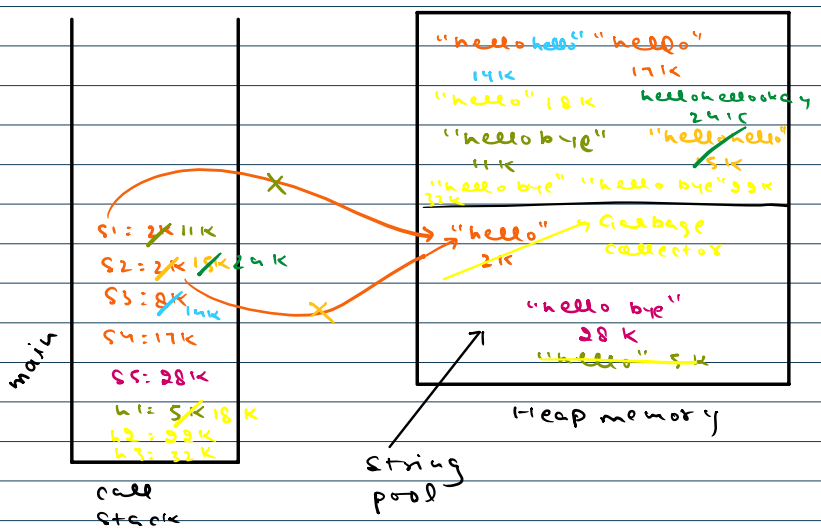


String concatenation

+ => to combine two or more than two strings.

what happens when we modify the string?

```
public class StringPoolDemo {
    public static void main(String args[]) {
        String s1 = "hello";
        String s2 = "hello";

        String s3 = new String( original: "hello");
        String s4 = new String( original: "hello");

        s1 = s1 + "bye";
        s2 = s2 + s3;
        s3 = s3+s4;

        s2 = s2.concat( str: "okay");

        String s = "hello" + "bye";

        String h1 = "hello";

        String h2 = h1 + new String( original: " bye");
        String h3 = h1 + new String( original: " bye");
        System.out.println(h2 == h3);
    }
}
```



String + [any data type] : String

charAt

String s : "Akaash"
0 1 2 3 4

s[2] : 'a'
s.charAt(2) : 'a'

compareTo() → comparing strings lexicographically.

s1.compareTo(s2);

0 → both strings are equal.
<0 → if s1 comes before s2 → s1: "Abhay" s2: "Akaash"
>0 → if s2 comes before s1

What is a substring?
It is a contiguous sequence of chars within a string.
word → "akash"

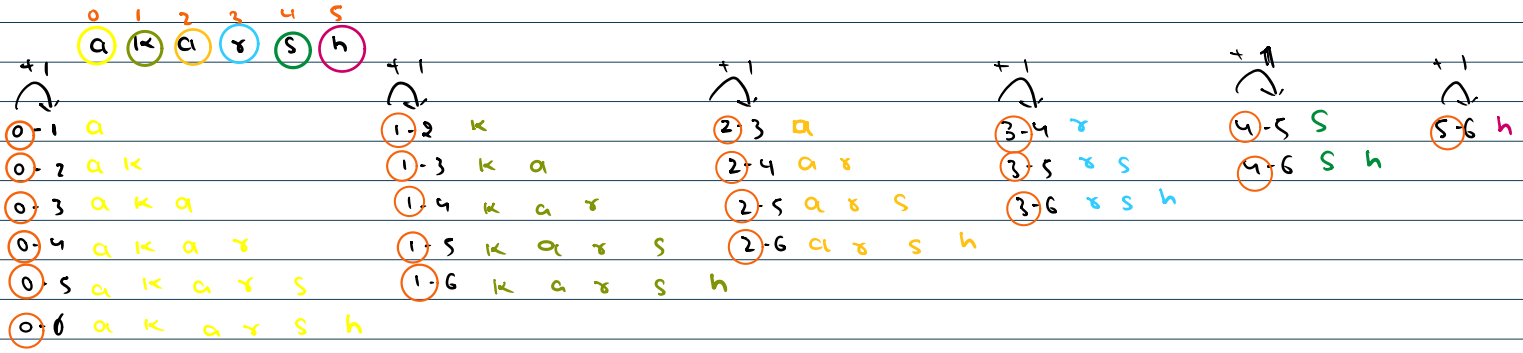
"ak" "ka" "as" "sh" "ash" "kash" "akash"

substring() => to extract a portion of a string.

s.substring(startIndex, endIndex);

inclusive
exclusive {optional}

Print all substrings




```
for (int i=0;i<s.length();i++) {
    for (int j=i+1;j<=s.length();j++) {
        s.substring(i,j);
    }
}
```

https://leetcode.com/problems/reverse-words-in-a-string/